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COP 4834 Web Systems II

Assignment 10: Adding and Ensuring Security

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Assignment:

For this assignment we will be adding a layer of security into the application. We will also include a documented layer to make sure that only those who are necessary should have access to the information inside the application.

Title: Adding and Ensuring Security

Objective:

To implement a security layer on your application

User Roles:

Admin: Create, Read, Update, Delete, Manage Roles

User: Create and Read

Access Control Plan:

This application uses a Role-Based Access Control model to manage user permissions. There are two roles that are defined: Admin and Regular User. Each role is assigned with specific permissions that determine what actions the user can perform within the system. This ensures that sensitive operations such as updating or deleting records are restricted to admin users only. The Admin role has full access to all CRUD operations and can also manage user roles. The Regular User role has limited access and can only perform Create and Read operations. Access control is enforced at both the user interface level and the server level to prevent any potential unauthorized actions.

ACTION	ADMIN	USER
Login	Yes	Yes
Create	Yes	Yes

Read	Yes	Yes
Update	Yes	No
Delete	Yes	No
Manage Roles	Yes	No

Login and Authentication:

Admin Log in-

Username: admin@test.com

Password: admin123

User Log in-

Username: user@test.com

Password: user123

Security Implementation:

The security layer is implemented by using a combination of authentication and authorized mechanisms. The authentication process is handled through the log in process by having the user enter their username and password. Authorization is handled through having the user verify their credentials before being able to make any actions. This helps verify that the user is only making the actions that their credentials allow them to. Therefore if a user attempts to make an action without the proper log in the system will deny them access. Another precaution will be that only certain actions will be visible to the user based on their log in. For example, the create and delete buttons will not be visible if the user is working under normal user credentials.

Role Management Interface:

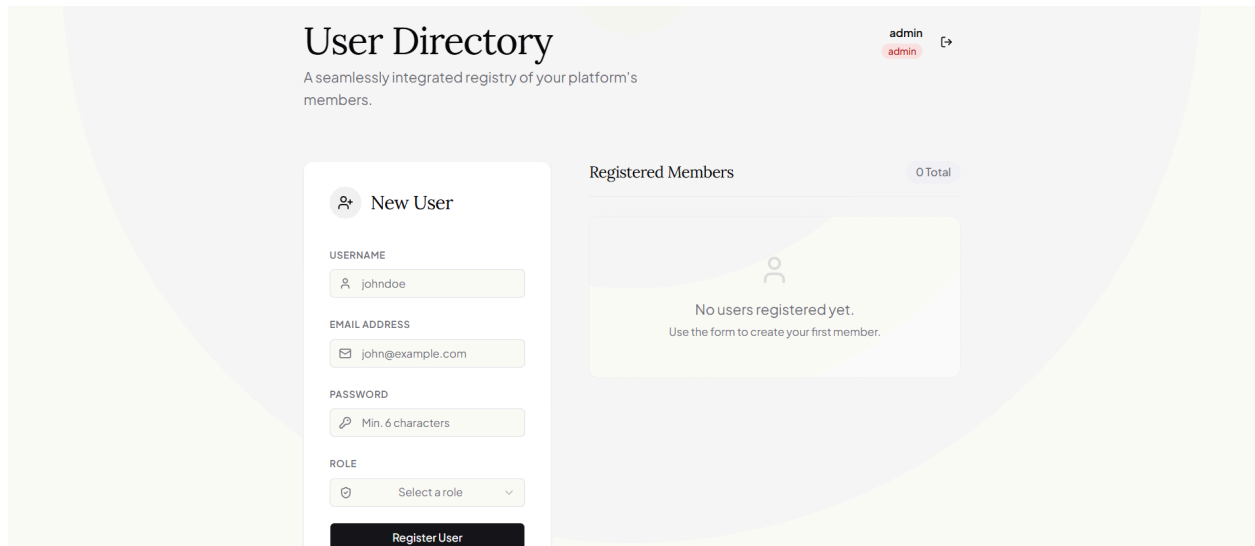
This application does include a role management interface that will allow the admin to view and manage any of the user roles. This interface will display a list of the users along with their role in the application. This will help provide clear communication to the admin about which users are permitted to do which actions.

Deployed Examples:

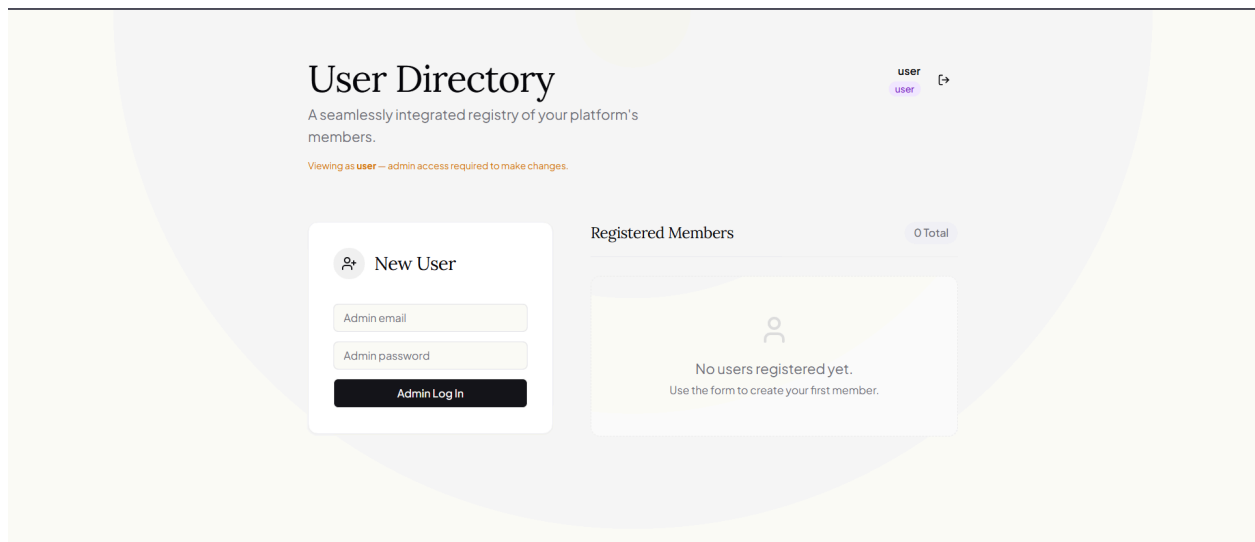
In the examples in the appendix below are examples of the admin log in and the regular user log in. As you can see a user is logged in under admin credentials they have access to add a new user and assign them a role. For another example whenever a regular user tries to log in through the admin section they will receive a “Login Failed” and they will not be able to log in. Also these create buttons will not be visible under a regular user credential.

Appendix A: Screenshots

Admin Log in success



User Log in success



User attempts to log in as admin

User Directory

A seamlessly integrated registry of your platform's members.

Log in with your email and password. Admin accounts can create, edit, or delete users.

 New User

Unexpected token '<', '<IDOCTYPE'...

is not valid JSON

Registered Members

0 Total



Log in to view members.

Enter your email and password to continue.

Login failed

Unexpected token '<', '<IDOCTYPE'...' is not valid JSON